

OpenClaw Plant Care Assistant Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw Jetson-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Solution Configuration Tests] in the `Pre-use Preparation` directory. If you are only using TUI / WhatsApp, you can skip `04` for now. This article assumes the basic environment is already prepared and only covers the plant care project itself.

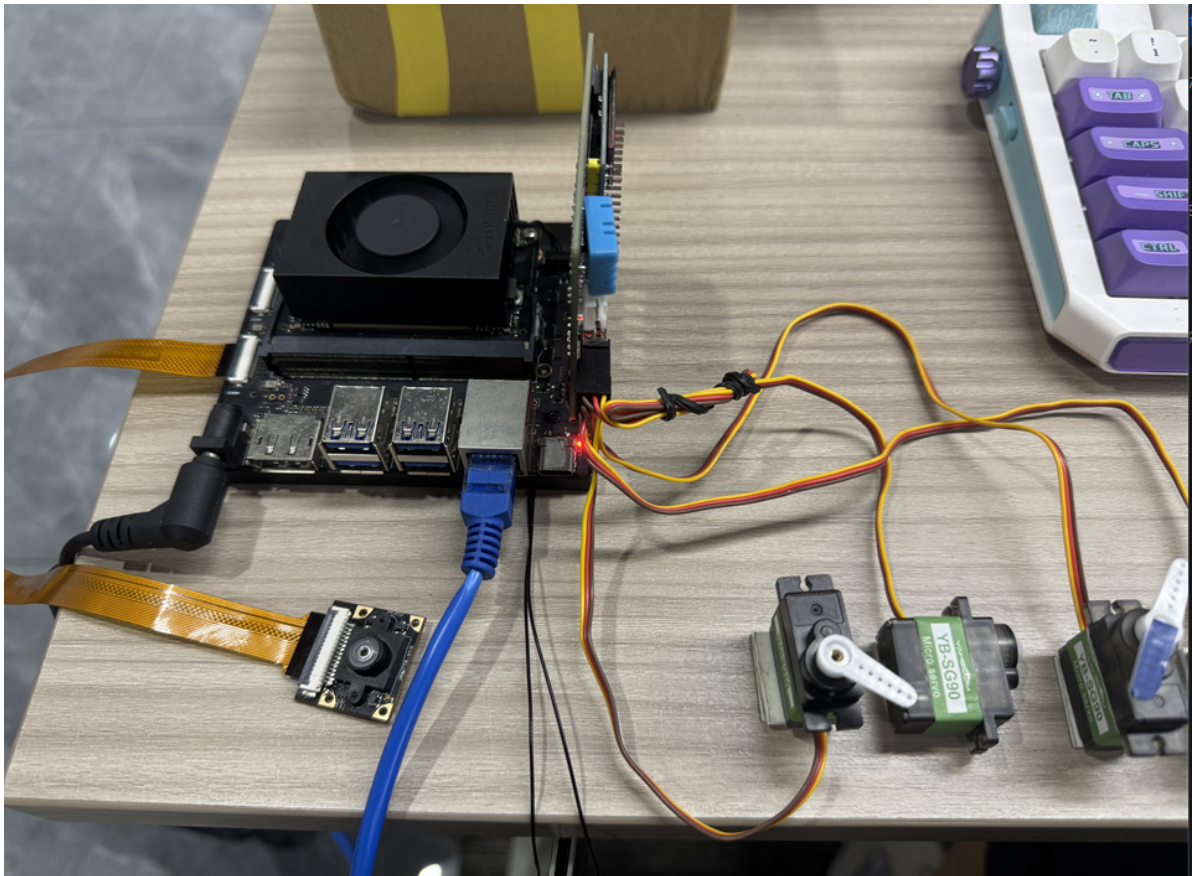
1. Tutorial Objective

This article demonstrates OpenClaw's multi-peripheral collaboration capability through a plant care scenario.

- Learn how to combine temperature/humidity sensors, OLED, RGB, and camera into a plant care assistant
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature/humidity sensor
- OLED display
- RGB light strip / RGB LED module
- CSI camera
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Camera capture command: `/home/jetson/hardware_hal_demo/seewhat`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project does not rely on a single "large project script"; instead, it has OpenClaw orchestrate multiple basic tools:

- `dht` is responsible for reading ambient temperature and humidity
- `oled` is responsible for displaying care information
- `rgb` is responsible for status alerts
- `seewhat` is responsible for observing plant appearance

4. Testing Basic Project Capabilities

Before starting this project, please complete an `openclaw tui` basic dialogue self-check first; if you haven't done so, see [05-Three Dialogue Solution Configuration Tests] first. After passing, use TUI to verify the basic capabilities in the project.

Enter in terminal:

```
openclaw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
  
session agent:main:main  
connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)  
|  
|
```

After confirming OpenClaw can communicate normally, you can try the following types of commands:

- Read current temperature and humidity
- Display "Plant OK" on the OLED
- Change the RGB light to green
- Take a photo for me

If all these individual capabilities work, it means the main components required for the project are all ready.

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
WhatsApp	Remote care, classroom demonstrations	Easy to set up; suitable for remote status monitoring	Demonstrate "remote natural language care"
TUI	Local debugging, project integration	Most direct, easiest to troubleshoot	First run through multi-peripheral links
Voice code	Voice companionship, complete AI experience	Closer to a real assistant experience	Use as final demonstration solution

If you have selected the AI voice module + speaker, Feishu dialogue, tui, and whatsapp dialogue need to play back OpenClaw's replies as audio. You can run the following program in the terminal. The AI voice module dialogue does not need to run this program repeatedly.

- Modify the TTS toggle parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to play audio output. After modification, save and run the following program in the terminal
```

- Enter in terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for the plant care project.

After successfully creating the bot, you can remotely inquire about plant status in WhatsApp, for example:

- Take a photo to check the current plant status and perform plant care

20:10



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current plant status and perform plant care

20:09 ✓✓



I can see the plant! It's a small potted plant with vibrant green-yellow leaves. The foliage looks full and healthy. Let me complete the plant care workflow.

20:10 ✓✓

Plant care check completed!

Status:

- 📸 **Visual check:** Plant looks healthy with vibrant green-yellow foliage
- 🌡️ **Temperature:** 29.7°C (85.5°F)
- 💧 **Humidity:** 52%
- 📺 **OLED:** Displaved "Plant:"



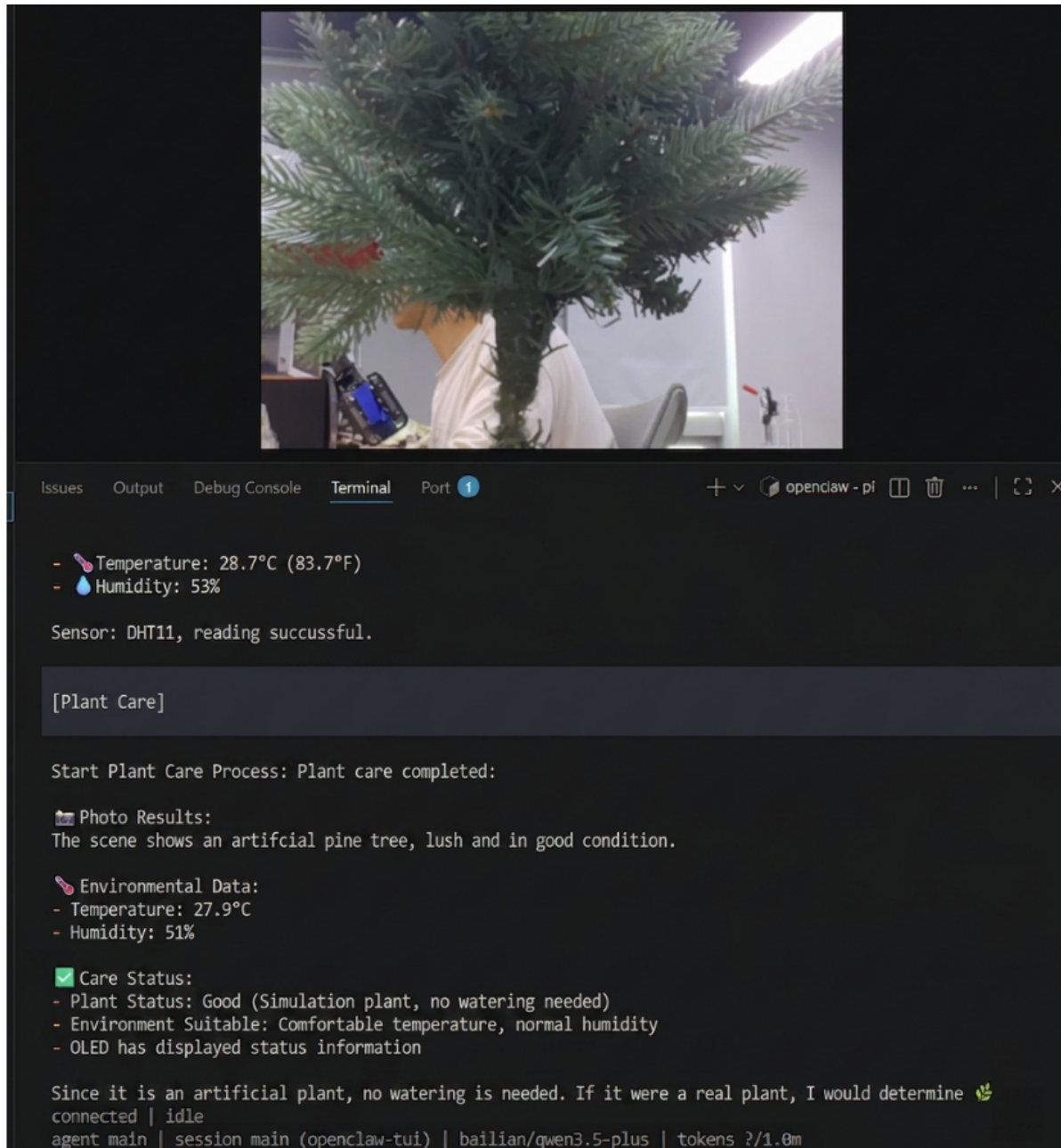
5.2 TUI Solution

If you have completed the TUI self-check in the Prerequisites section, directly enter terminal text dialogue:

```
openclaw tui
```

Then you can directly input project-related commands, for example:

- Plant care



5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [04-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- Check the plant status for me, is now a good time to water

```
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Me check the plant status and whether it's suitable for watering now.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me analyze the plant's condition and check the environment to
          determine if it needs watering.

## 📷 Plant Status Check
          **Environmental Conditions:**
- **Temperature:** 30.1°C (86.2°F) -
          Quite warm! 🌡️
- **Humidity:** 52% - Moderate

**Plant Analysis:**
          Looking at the captured image, I notice:
- The plant has **bright
          green/yellow foliage** with uniform coloring
- Leaves appear
          **vibrant and upright** (no wilting visible)
- The foliage has an
          **artificial appearance** (plastic-like sheen)

**🕒 Watering
          Recommendation:**
```

6. Project Implementation

Note: The following demonstrations are all conducted using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Scenario 1: Environment Reading

Experiment Objective

Read the current temperature and humidity around the plant to enable basic environmental monitoring.

Effect Description

OpenClaw will call DHT to read current environmental data and use natural language to report the temperature and humidity.

Example Commands

- Read the current temperature and humidity near the plant
- What are the current ambient temperature and humidity?
- Help me check if the potted plant area is too dry

6.2 Scenario 2: Status Judgment

Experiment Objective

Have OpenClaw provide a brief plant care suggestion after obtaining environmental data.

Effect Description

In addition to directly reporting values, OpenClaw can also summarize conclusions like "too dry", "too hot", or "normal status" in a more everyday language style.

Example Commands

- Judge the plant status based on current temperature and humidity
- Is now a good time to water

6.3 Scenario 3: Audio-Visual Alerts

Experiment Objective

When environmental conditions are not ideal, use OLED and RGB to provide more intuitive alerts.

Effect Description

For example, light green when the environment is normal, yellow when too hot, red when abnormal; simultaneously display simple prompt messages on the OLED, so it's not just "talking" but also has clear peripheral feedback.

Example Commands

- Light green if the environment is normal
- Light yellow and alert me if it's too hot
- Display "Plant Needs Care" on the OLED

Extensible Play Ideas

- Integrate with a buzzer for stronger alerts
- Make it into a desktop care status light
- Add timed polling and auto-refresh

6.4 Scenario 4: Photo Observation

Experiment Objective

Use the camera to capture the current plant image and have OpenClaw provide a simple description.

Effect Description

This step is suitable for observing leaves, pot surfaces, or overall growth. The current tutorial focuses on "take a look and describe", not automatic disease detection, making it more suitable for teaching demonstrations and basic visual monitoring.

Example Commands

- Take a photo of the plant
- Check the leaf condition
- Describe the current plant image after taking a photo